Interpretation of Models on the Onset of Diabetes using Machine Learning

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Abstract-- Diabetes mellitus is one of the most serious health challenges faced by the world today. The objective is to diagnostically predict whether or not a patient has diabetes, based on certain diagnostic parameters included in the dataset. Under the supervised learning, we have used Logistic Regression and Decision Tree. Using Decision Tree, the accuracy came out to be approximately 69%. In order to further test the accuracy, Logistic Regression predicted the accuracy to be 78%. The prediction is further done and graphically represented through K-Means and R-Squared Method. The test error and accuracy for logistic regression are then calculated using Confusion Matrix. The performance of classifier is evaluated based on precision, recall, accuracy and is estimated over correct and incorrect instances.

Keywords-- Diabetes Mellitus, Machine Learning, Logistic Regression, Decision Tree, K-Means.

I. INTRODUCTION

Diabetes mellitus is a kind of metabolic diseases in which a person has high blood sugar. There are two general reasons for diabetes: 1- the body does not produce enough insulin. Only 5-10% of people with diabetes have this form of the disease(type1). With the help of insulin therapy and other treatments, even young children with type1 diabetes can learn to manage their condition and live healthy. 2- cells do not respond to the insulin that is produced (type2). Insulin is the most essential hormone that regulates uptake of glucose from the blood into most cells (not central nervous system cells). Thus, deficiency of insulin plays a central role in all forms of diabetes mellitus. Type 2 diabetes mellitus is a complex disease of major health importance. This incidence is rapidly increasing in the developed countries. It is estimated by the year 2030, there will be 366 million people affected by Type 2 diabetes worldwide. Medical diagnostics is quite difficult task which is mostly done by expert doctors. An expert doctor commonly takes decisions by evaluating the current test results of a patient or compares the patient's results with other patients with the same condition by referring to the previous decisions. Thus, it is very difficult for a physician to diagnose hepatitis. For this reason, in recent times, many machine learning and data mining techniques have been considered to design automatic diagnosis system for diabetes.

Machine learning is an upcoming new technology with a wide range of applications. It has the potential to become one of the key components of intelligent information systems, inferred from large databases of recorded information that is to be applied as knowledge in various practical ways—such as being embedded in automatic processes, or used directly for communicating with human experts and for educational purposes. Most of the research work is directed towards the invention of new algorithms for learning, rather than towards gaining experience in applying existing techniques to real problems.

II. RELATED WORK

Design of the prediction models for diabetes diagnosis has been an area of active research for the past decade. Most of the models found are based on clustering algorithms. In the proposed project the employed three techniques namely are: Regression algorithm, K-means clustering and Classification algorithm, for the classification of the patients suffering from diabetes mellitus. The performance for K-means proved to be better than others when all the similar symptoms were grouped into clusters using these algorithms. In recent times, a study conducted is intended to discover the hidden knowledge from a particular dataset to improve the quality of healthcare for the diabetic patients. This paper approaches to aim the diagnosis of diabetes by using Machine Learning and demonstrated the need for pre-processing and replacing missing values in the dataset. Through the learned training set, a better accuracy was achieved with lesser time. Hence, there is a requirement of a model that can be developed that provides reliable, faster and cost effective methods to provide information related to the probability of patients suffering from diabetes. In the present work, an attempt is made to analyse various diabetes parameters and to establish a probabilistic relation between them using Machine Learning Models.

III. PROPOSED METHODOLOGY

A. Logistic Regression

Logistic regression is a applied mathematics model that in its basic type uses a logistical operate to model a binary variable. In multivariate analysis, logistical regression (or logit regression) is estimating the parameters of a logistical model (a type of binary regression). Mathematically, a binary logistical model includes a variable with 2 attainable values, like pass/fail that is portrayed by associate indicator variable, wherever the 2 values area unit labelled "0" and "1". within the logistical model, the log-odds (the exponent of the odds) for the worth labelled "1" may be a linear combination of 1 or a lot of freelance variables ("predictors"); the freelance variables will every be a binary variable (two categories, coded by associate indicator variable) or a nonstop variable (any real value). The corresponding likelihood of the worth labelled "1" will vary between zero (certainly the worth "0") and one (certainly the worth "1"), hence the labelling; the operate that converts log-odds to likelihood is that the logistical operate, thence the name.

The equation utilized in logistical Regression is:

$p/1-p = \exp(b0 + b1x)$

B. Decision Trees

Decision tree is a tree structure, that is within the type of a flow chart. It's used as a technique for classification and prediction with illustration exploitation nodes and internodes. the foundation and internal nodes area unit the take a look at cases that area unit won't to separate the instances with totally different options. Internal nodes themselves area unit the results of attribute take a look at cases. Leaf nodes denote the category variable. Figure 1. shows a sample call tree structure.

International Journal of information Mining & information Management method (IJDKP) Vol.5, No.1, Gregorian calendar month 2015 five Figure one. Sample call Tree Structure call tree provides a robust technique for classification and prediction in polygenic disorder designation drawback. varied call algorithms area unit on the market to classify the information, together with ID3, C4.5, C5, J48, CART and CHAID. during this paper, J48 call tree algorithmic rule [10] has been chosen to determine the model. every node for the choice tree is found by scheming the very best info gain for all attributes associated if a particular attribute provides an unambiguous end result (explicit classification of sophistication attribute), the branch of this attribute is terminated and target worth is allotted thereto.

IV. FLOWCHART

The following fig. 1 depicts a general methodology or the flow of diabetes prediction system. we propose to predict whether a patient will suffer from diabetes or not in the future. In this process we will first collect a database of diabetes from hospitals or laboratories. Then we will be categorizing the collected data into diabetic patient and non-diabetic patient. We then extract the features from the dataset. From these features we select the attributes which are more responsible for diabetes in a patient. This stage is Feature selection. Later, we pre-process these features, to make it easier to process them and also to increase the accuracy. This dataset is then split into training dataset and testing dataset. The train-test split is of the ratio 70:30 which means that 70% of the data will be our training data and the rest 30% will be our testing data. With the help of feature selection, we are able to get precise accuracy. This output is then evaluated using performance metrics. Machine learning models are used and the one with the better accuracy will be chosen to further train the model.

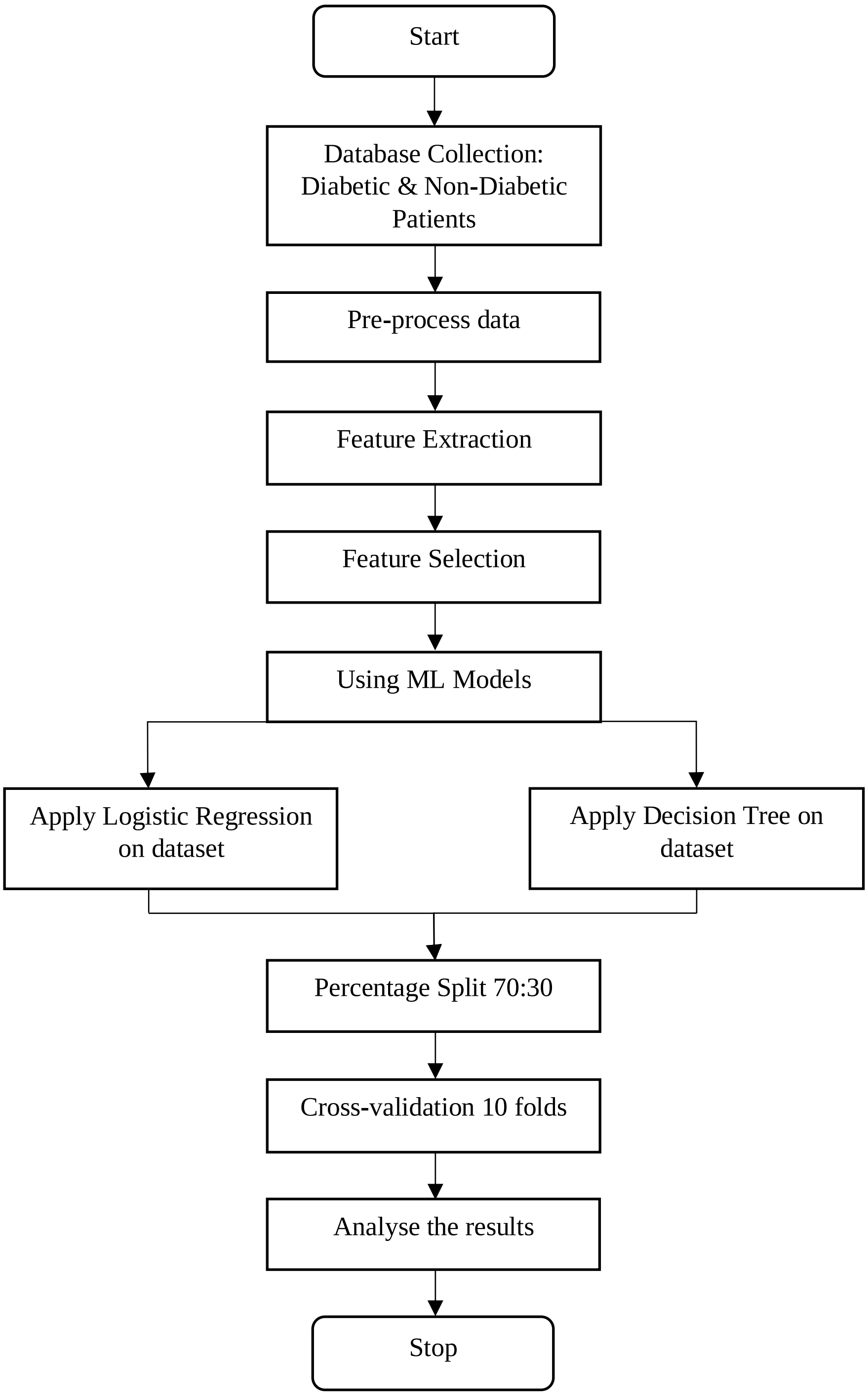


Fig 1. Flowchart of Diabetes Prediction System

V. DATASET DESCRIPTION AND PRE-PROCESSING

The paper explores the aspect of Decision Tree and Logistic Regression as Data Mining techniques in determining diabetes in women. The classification type of data mining technique has been applied to the Pima Indians Diabetes Dataset of National Institute of Diabetes and Digestive and Kidney Disease as shown in Table 1. It contains 8 n

TABLE 1

Dataset Description

Dataset	No. of Attributes	No. of Instances
Pima Indians Diabetes Database of National Institute of Diabetes and Digestive and Kidney Diseases	8	768

TABLE 2

Attribute Description

Attribute	Re-labeled values
1.Number of times Pregnant	Preg
2.Plasma Glucose concentration	Plas
3.Diastolic blood Pressure (mm Hg)	Pres
4.Triceps skin fold thickness (mm)	Skin
5.2-Hour Serum Insulin	Insu
6. Body Mass index (kg/mm^2)	Mass
7.Diabetes Pedigree function	Pedi
8. Age (Years)	Age
9. Class Variable (0 or 1)	Class

Transformation steps include:

- Replacing missing values, and
- Normalization of values.

The latter makes it easy to use the dataset, as the range of the variables are 0 and 1.The Table 2 depicts significant attributes obtained after execution are as follows:

1) Plasma glucose concentration (Plas)
2) Body mass index (kg/m2) (Mass)
3) Diabetes pedigree function (Pedi)
4) Age (years)
5) Class Variable (nominal)-determines whether the person is suffering from diabetes or not.

After pre-processing, it is found that 130 patients were suffering from diabetes (Positive) and 262 patients were not (Negative). It is graphically represented in the fig.2.

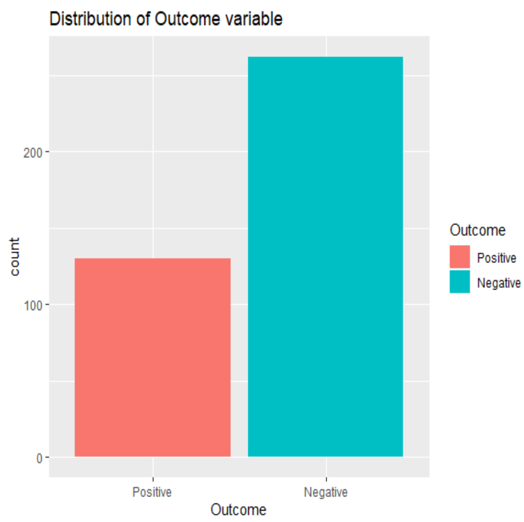


Fig. 2 distribution of the Outcome

VI. IMPLEMENTATION

In this paper two algorithms namely, J48 (decision tree algorithm) and Logistic Regression, have been used to create the model for diagnosis. The dataset was divided into training dataset and testing dataset by the cross-validation technique and percentage split technique.

10-fold cross validation is used to prepare training and testing dataset. After data pre-processing (CSV format), the Logistic algorithm is employed on the dataset after which data are divided into "tested-positive" or "tested-negative" depending on the final result of the decision tree that is constructed. The algorithm is as follows:

A. Algorithm

Input:

- Pima Indians Diabetes Dataset of National Institute of Diabetes and Digestive and Kidney Diseases dataset pre-processed in CSV format.

Output:

- J48 Decision Tree Predictive Model with leaf node either tested-positive or tested negative and Logistic Regression Prediction Results.

(1) The dataset is pre-processed. The further operations are performed on the dataset are:

(i) Replace Missing Values and

(ii) Normalization of values.

(2) Processed dataset is passed through feature selection wherein certain attributes are deleted from the dataset.

(3) The final processed dataset is uploaded in csv file.

(4) The J48 Decision Tree and Logistic Regression algorithm are employed.

(5) For purposes of the algorithms, Cross-Validation and Percentage Split techniques are applied for model validation.

B. Software Application Approach

With the advent of smartphones, patients are increasingly using mobile technology through automated text messages and various applications, or "apps," to monitor their disease states. In 2010, an estimated 10.9 million people > 65 years of age and 215,000 people < 20 years of age in the United States had diabetes.

In the fast-paced and evolving world of technology, younger populations are often highly proficient with and adaptable to smartphones. As a result, mobile apps for Apple iOS and Android operating systems have overwhelming popularity. More than 500 million Apple iOS and Android devices have been activated since 2007.

In fig.3 the system architecture of diabetes prediction system is shown. We have trained diabetes dataset using Machine learning models. This software application is beneficial for both non-diabetic and diabetic patients. While training the model, age is one of the important factors for predicting diabetes. Hence this application can be used by any age group. For the people not suffering from diabetes (based on age, weight and height) can also use this application in order to predict if they will be prone to diabetes in the near future. The people suffering from diabetes can keep track of the factors such as blood pressure, insulin levels, glucose etc on a daily basis. The application aims to prevent the onset of diabetes by forecasting the disease earlier and the precautions could be taken at an early stage.

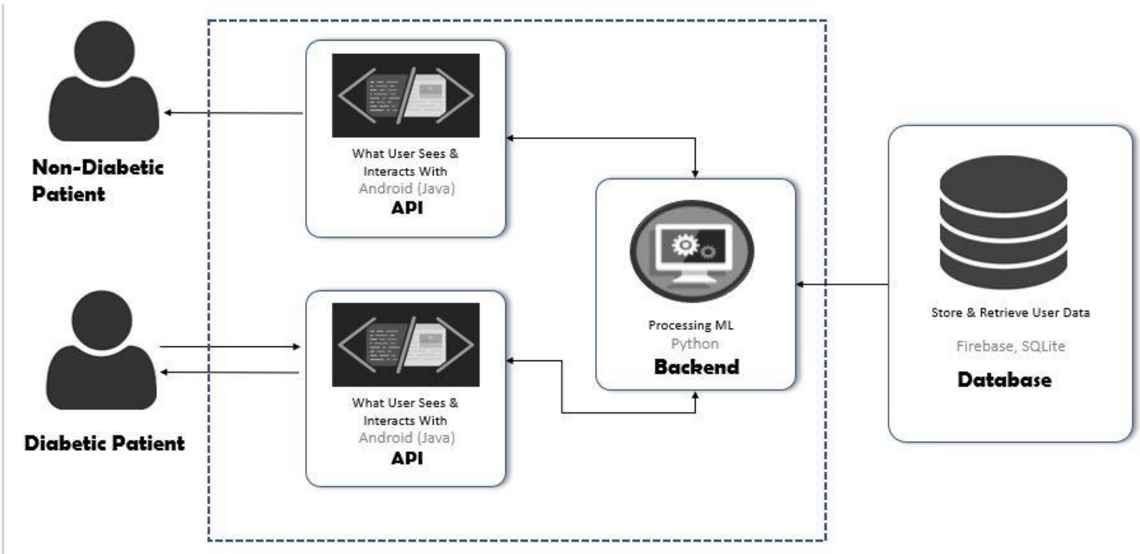


Figure. 3 System Architecture of Diabetes Prediction System

VII. RESULT AND ANALYSIS

A. Logistic Regression

1) Based on Percentage Split (70:30) Technique: Since a 70:30 percentage split was applied on the dataset 119 of the instances were used as the test dataset while the rest of were using for training the model. The logistic algorithm gives the following correctness results for the given dataset. The Table 3 shows the number of True-Positives (TP), True-Negatives (TN), False-Positives(FP) and False-Negatives(FN) in a Confusion Matrix.

TABLE 3

Confusion Matrix and Statistics

	Prediction	
Actual	0	1
0	71	17
1	8	22

TABLE 4

Performance Measures from Logistic Regression

Accuracy	0.7881
95% CI	(0.7033,0.858)
No Information Rate	0.6695
P-Value[Acc>NIR]	0.003185
Kappa	0.4916
Mcnemar’s Test P-Value	0.109599
Sensitivity	0.8987
Specificity	0.5641
Positive Pred Value	0.8068
Negative Pred Value	0.7333
Prevalence	0.6695
Detection Rate	0.6017
Detection Prevalence	0.7458
Balanced Accuracy	0.7314
‘Positive’ Class	0

2)Terminologies of Test Statistics

The above given table depicts Performance Measures from Logistic Regression. These performance measures are calculated with the help of confusion matrix. These values show the correctness of results achieved from Logistic Regression.

1) Kappa Statistic: It is a metric that compares the Observed Accuracy with Expected Accuracy.

2) Mcnemar's Test P-Value: It is a statistical test for paired nominal data. In context of machine learning models, we can use McNemar's Test to compare the predictive accuracy of the two models.

3) Sensitivity (also called the true positive rate, the recall, or probability of detection in some fields) measures the proportion of actual positives that are correctly identified.

4)Specificity (also called the true negative rate) measures the proportion of actual negatives that are correctly identified.

5)Mean Absolute Error: average of the absolute error between observed and predicted value.

6)Root Mean Squared Error: It measure of the differences between value (Sample and population values) predicted by a model or an estimator and the values actually observed

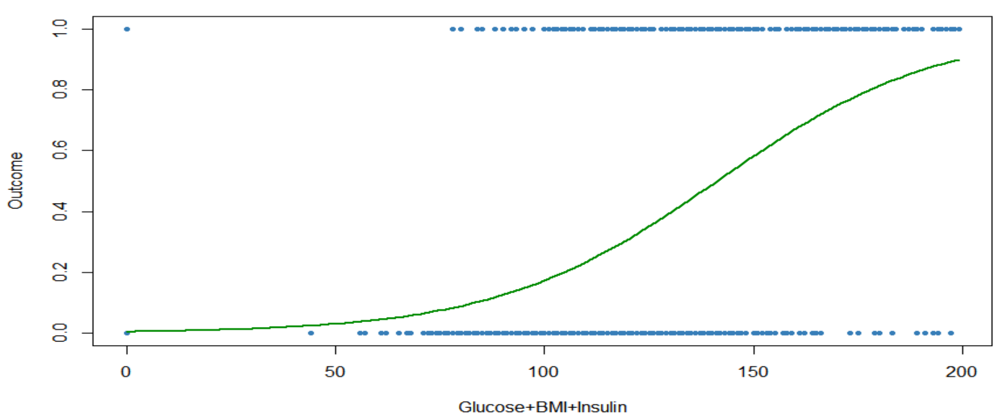
7)Relative Absolute Error: It is the ratio of the absolute error of the measurement to the accepted measurement.

TABLE 5

Confusion Matrix of Logistic Regression

	Prediction	
Actual	0	1
0	233	5
1	29	74

In the table 5, the values represent following:
i. Number of correct outcome that the instance tested positive
ii. Number of incorrect outcome that the instance tested negative
iii. Number of incorrect outcome that the instance tested positive
iv. Number of correct outcome that the instance tested negative



B. Decision Tree

Decision tree J48 implements Quinlan's C4.5 algorithm [5] for generating pruned tree. The tree generated can be used for classification of a patient that has being tested positive or negative for diabetes. The data mining technique uses the thought of information gain. Each attribute of the data is used to make a decision by splitting the data into smaller or sub-modules.

It examines normalized information gain (difference in entropy) that results from choosing an attribute as a split point. The highest normalized IG is used at the root of the decision tree. This procedure is repeated until the leaf node is created for the tree specifying the class attribute that is chosen. From the results obtained, both the methods have a comparatively small difference in error rate, though the percentage split of 70:30 for Logistic Regression technique gives the least error rate compared to other two decision tree implementations. The machine learning models are efficient in the diagnosis of diabetes using the percentage split of 70:30 of the data set. A developed model for diagnosis of diabetes will require more training data for creation and testing which would be more beneficial for our software application.

C. Performance Analysis

Performance of the model is being evaluated on the basis of ROC curve. ROC Curve stands for Receiver Operating Characteristics Curve which is represented below in the fig,7. The threshold value we took for analysing the performance is 0.5.

By using both the supervised learning models, it is known that the Logistic regression model works better on classified or labelled data. The accuracy of the Logistic Regression came out to be 0.7884 whereas that of Decision Tree came out to be 0.6933. In fig. 6 we can easily predict that the Logistic Regression Model gives better performance and less error rate than Decision Tree. This result will be helpful in order to achieve good results for real time datasets.

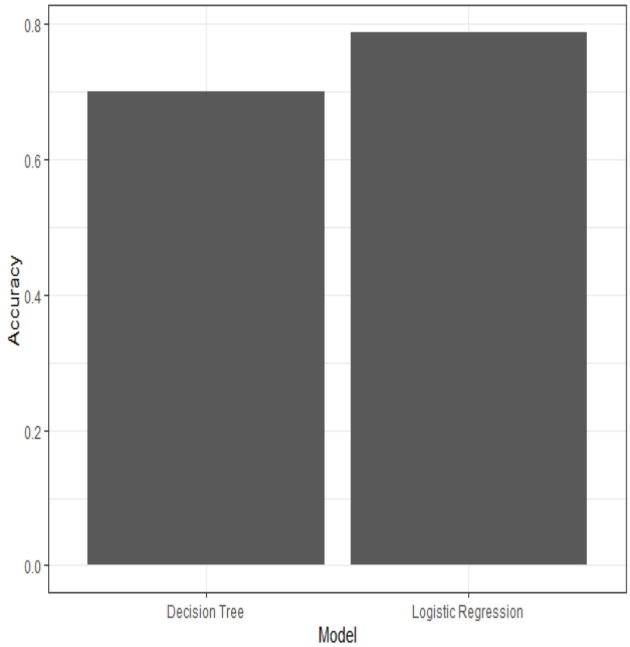


Fig. 6 Comparison of Model Accuracy

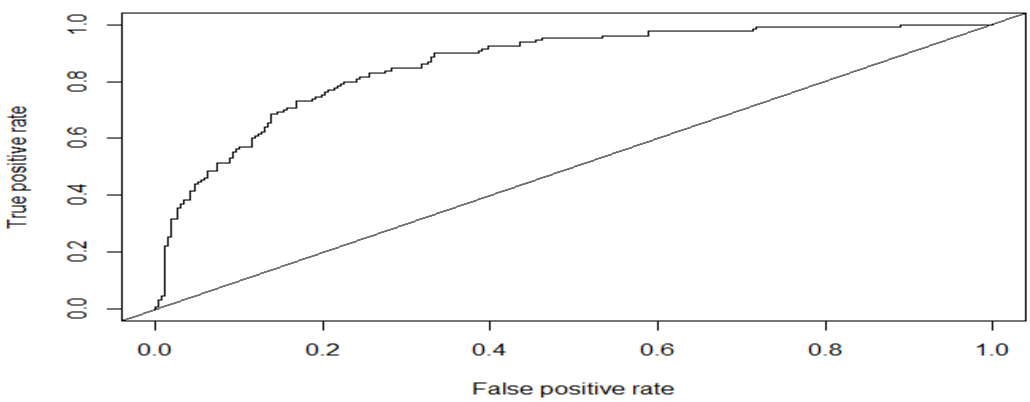


Fig.7 ROC Curve

VII. CONCLUSION

The automatic prediction of diabetes is an important real world medical problem. Detection of diabetes in its early stages is the successful key for treatment. This paper shows how the Data Mining techniques are used to model actual diagnosis of diabetes for local and systematic treatment, along with presenting related work in the field. Experimental results show the effectiveness of the projected model. The performance of the techniques was investigated for the diabetes designation downside. Experimental results demonstrate the acceptableness of the projected model. In future, it's planned to assemble additional data from completely different locales everywhere the globe and build a additional precise and general discerning model for diabetes conclusion. Future study can likewise specialize in gathering data from a later period and see new potential prognostic parts to be incorporated. The work may be extended and improved for the automation of diabetes analysis.

VIII. FUTURE SCOPE

There is so much to learn in this vast field of data science. In our project, we have only focused on one set of population living in a particular region. Our aim is to expand our work on a city, where we would add more attributes responsible for diabetes. This would be a good estimation for the onset of diabetes. This would alert people and could take precautions beforehand. We aim to prepare a website where people would enter their details and would predict whether they would be prone to such disease called diabetes in the near future.

IX. REFERENCES

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